First unit indices minimize both index sum and residue energy

p	first 4 unit indices	$S_{p,4}$	$E_{p,4}$	v for e=1
2	[1, 3, 5, 7]	16	8	32
3	[1, 2, 4, 5]	12	2	16
5	[1, 2, 3, 4]	10	0	10

$$v_p(H_{r,p}(t)) = e\, S_{p,r} + 2\, E_{p,r}$$